

PRIMITIVES CANON

2026-03-18

Dexter Hadley, MD/PhD

Hadley Lab CANONIC

Abstract

INTEL, COIN, and TALK are the three native primitives of the GALAXY operating surface.

hadleylab.org Governed Research. Every claim cited.

Contents

1

Constraints

2

Primitive Composition

1. Constraints

- 1

MUST:

INTEL pane renders scope intelligence (summary)
- 1

MUST:

INTEL pane is actionable (coverage gaps show)
- 1

MUST:

COIN bar shows wallet balance in control pane
- 1

MUST:

COIN bar supports transaction feed (mints, swaps)
- 1

MUST:

COIN bar shows scope-level economy when viewed
- 1

MUST:

TALK interface replaces search dock at bottom
- 1

MUST:

TALK supports dual mode: search (default) and edit
- 1

MUST:

TALK carries current scope context in every message
- 1

MUST:

TALK streams responses via SSE in expandable list
- 1

MUST:

TALK recognizes governance commands (create, delete, update)
- 1

MUST:

All three primitives update contextually on demand
- 1

MUST:

TALK responses that reference scopes are clickable
- 1

MUST:

INTEL coverage gap "fix" button opens TALK with prompt
- 1

MUST:

Every TALK edit operation logs to COIN ledger
- 1

MUST:

Every action attributed to authenticated GitHub user
- 1

MUST NOT:

Render primitives as separate pages (they co-exist)
- 1

MUST NOT:

Allow TALK editing without authenticated session
- 1

MUST NOT:

Bypass build pipeline for governance mutations

2. Primitive Composition

- Navigate to scope

INTEL updates (that scope's intelligence)

COIN updates (that scope's economy)

TALK context updates (that scope's context)
- TALK edit command

Worker executes governed operation

COIN ledger records CONTRIBUTE event

Build pipeline triggers galaxy regeneration

Frontend hot-reloads with new scope
- INTEL gap found

"Fix" button opens TALK with prompt

User describes fix in conversation

TALK creates governance artifact

Gap closes on next build